

Trishit Swarnakar

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[Linked In](#) | [GitHub](#) | [LeetCode](#) | [Portfolio](#)



SUMMARY

Computer Science and AI undergraduate working across software engineering, applied AI, and product development. Build full-stack applications, RAG systems, and agent workflows; experience with LLM evaluations, model orchestration, and real-time collaboration.

EDUCATION

B.Tech, Computer Science & Artificial Intelligence (Minor in Finance) 2024 – 2028
Newton School of Technology, Rishihood University

Intermediate (XII) – Kalyani Central Model School (84.4%); **Matriculation (X)** – Springdale High School (87.0%)

EXPERIENCE

LLM Research Engineer – Aicoo Jun 2026 – Sep 2026
Remote – Oxford, United Kingdom Internship

- Built evaluation harnesses to test LLM behavior, prompt-injection defenses, and output guardrails.
- Prototyped retrieval and agent components; evaluated accuracy and robustness to guide implementation changes.
- Contributed to shared agent infrastructure and agent-to-agent decision-making.

PROJECTS

RYOKU – AI Customer-Support Platform [\[GitHub\]](#) [\[Demo\]](#) Feb 2026
Next.js, TypeScript, PostgreSQL, pgvector, Drizzle ORM, Redis, Pusher, Vercel AI SDK

- Built a RAG pipeline for PDF, DOCX, and web content using chunking, batch embedding with Google’s gemini-embedding-001, and thresholded top-k vector retrieval in PostgreSQL.
- Implemented Groq model rotation and API-key failover, streamed tool calls for escalation, lead capture, and web search, and added input sanitization, prompt-injection checks, and token limits.
- Built an agent workspace with Pusher pub/sub and Redis API rate limiting for live escalation queues, agent presence, and sentiment-aware handoff to human support agents.

VFLOW – Agentic AI Workflow Builder [\[GitHub\]](#) [\[Demo\]](#) Apr 2026
Next.js, TypeScript, Vercel AI SDK, React Flow, Zustand, Zod

- Built a visual workflow editor and execution engine that passes upstream node outputs into downstream steps, with streaming responses and AbortController-based cancellation.
- Integrated six model providers through a shared interface: OpenAI, Anthropic, Google, Groq, xAI, and OpenRouter. Added schema validation, browser-persisted workflows, and user-supplied API keys.

LLD Practice – AI-Assisted Design Practice [\[GitHub\]](#) Sep 2026
Next.js, TypeScript, Groq SDK, Zod, Zustand, Vitest, Playwright

- Built a low-level design practice app with guided worksheets, generated UML, and deterministic quiz scoring across Java, Python, and C++ variants; added browser-local submissions and revision history.
- Implemented rubric-based AI design reviews with schema validation and evidence checks against submitted fields, rejecting invented references and unsupported quotes before publishing feedback.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, SQL; embedded C/C++
Core CS:	Data Structures and Algorithms (DSA), Object-Oriented Programming, DBMS, Computer Networks, System Design, Concurrency, Real-Time Systems
AI / Agents:	LLMs, RAG, Agentic AI, Multi-Model Orchestration, Agent-to-Agent Communication, Model Context Protocol (MCP), Agent Skills and Plugins, AI/Agent Harnesses, LLM Evaluations, Prompt Engineering, Tool Calling, Guardrails, LangChain, LangGraph
ML / Data Science:	Machine Learning, Deep Learning, NLP, TensorFlow, scikit-learn, Hugging Face, NumPy, Pandas, SciPy
Full Stack / Data:	React, Next.js, React Native, Node.js, Express.js, Tailwind CSS, Vercel AI SDK, REST APIs, OAuth 2.0, PostgreSQL, MySQL, MongoDB, Redis, pgvector, Drizzle ORM
Networks / Cloud:	TCP/IP, HTTP/HTTPS, TLS, Client–Server and Peer-to-Peer Networking; AWS VPC, EC2, Security Groups, Ports, Route Tables, Gateways, Cloudflare Tunnel (coursework/labs)
Tools / Product:	Git, GitHub Actions, Jest, Swagger, Figma, n8n, Agile, Scrum

LEADERSHIP & EXTRA-CURRICULAR

Robotics & Hardware:	Technical Specialist, SAST (RIT Rover Innovation Team) and former Vice President, Robotics Club (RU × NST). Built ESP32/Arduino subsystems in C/C++, led CAD and robotics workshops, and contributed to teams competing at IIT Delhi, DTU, and SRM. Represented the club at Bharat Mandapam (Rishihood 10th Convocation).
Harvard HPAIR:	Delegate at the Harvard and Asia HPAIR conferences.
Beyond Tech:	Event Coordinator, Damru: led Battle of Bands (10 teams); lead guitarist (10+ years); co-founded a marketing agency and managed content for GoForGold and Life@Rishihood.